

Covariance and Causality

$$\begin{aligned}
 X_t &:= a_x X_{t-1} + a_{yx} Y_t + a_{zx} Z_t + \xi_t^x \\
 Z_t &:= a_z Z_{t-1} + a_{xz} X_{t-1} + a_{yz} Y_t + a_{wz} W_{t-1} + \xi_t^z \\
 Y_t &:= a_y Y_{t-1} + a_{xy} X_{t-1} + a_{wy} W_{t-2} + \xi_t^y \\
 W_t &:= a_w W_{t-1} + a_{yw} Y_t + a_{zw} Z_t + \xi_t^w \\
 U_t &:= a_u U_{t-1} + a_{wu} W_t + \xi_t^u.
 \end{aligned} \tag{1}$$

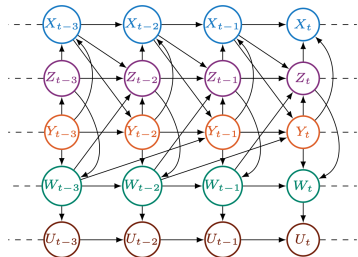


Figure 1: Running example. Left: Dynamic structural causal model (dynamic SCM). Right: Associated full-time causal graph $\mathcal{G}^f$ (FTCG).

Summary Causal Graph

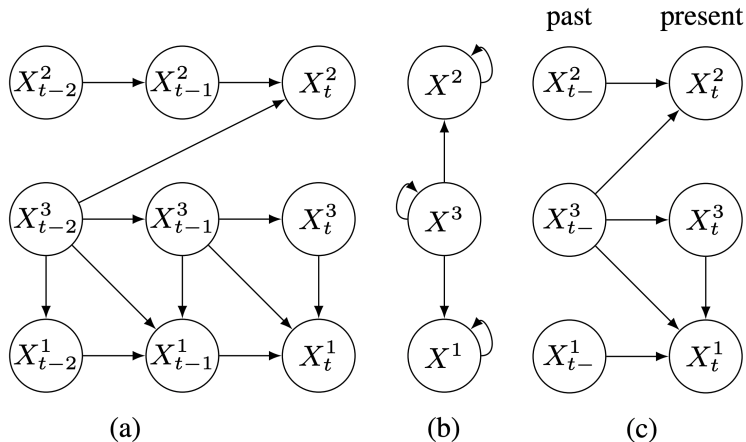


Figure 1: Example of a window causal graph (a), a summary causal graph (b) and an extended summary causal graph (c).

The dimensionality of the model

$$\begin{pmatrix} w_{1,1} & w_{1,2} & \cdots & w_{1,N} \\ w_{2,1} & w_{2,2} & \cdots & w_{2,N} \\ \vdots & \vdots & w_{K,T} & \vdots \\ w_{K,1} & w_{K,2} & \cdots & w_{K,N} \end{pmatrix} \underbrace{\begin{pmatrix} x_{t-1} & x_{t-2} & \cdots & x_{t-(N+T)} \\ x_{t-2} & x_{t-3} & \cdots & x_{t-(N+T-1)} \\ \vdots & \vdots & x_{t-T} & \vdots \\ x_{t-N} & x_{t-(N-1)} & \cdots & x_T \end{pmatrix}}_{\text{Hankel matrix } X_{t-1}}.$$

Convolution of f and g

$$(f * g)(t) = \int_{\tau=0}^T f(\tau) g(t - \tau) d\tau$$

The bilateral Laplace transform

$$F(s) \cdot G(s) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-st} f(u) g(t-u) du dt = \int_{-\infty}^{\infty} e^{-st} (f * g)(t) dt.$$

$$KL(p(w|X) || p(w|do(X=x)))$$

$$do(X_{t-p+1}=x_0)$$

$$(X_{t-p+1}, X_{t-p+2}, \dots, X_{t-1}, X_t) \rightarrow Y = X_{t+1}$$

sample (training data)

$$S^1: (\quad) \quad (\quad)$$

$$S^2: (\quad) \quad (\quad)$$

$$\vdots$$

$$S^n: (\quad) \quad (\quad)$$

model. $Y = \sigma(WX)$

$$W = \begin{bmatrix} W_1^T \\ \vdots \\ W_n^T \end{bmatrix}$$

$$W \in \mathbb{R}^{d \times n}, \quad X \in \mathbb{R}^{n \times 1}, \quad W_i \in \mathbb{R}^n$$

$$? P(W|X)$$

$$? P(W|do(X=x_0))$$

$$KL(P(W|X) || P(W|do(X=x_0)))$$